Draft version November 13, 2026
Typeset using LaTeX **twocolumn** style in AASTeX7.0.1

Detecting Electron Beam Heating in Active Region Moss with the Second Flight of the EUV Snapshot Imaging Spectrograph

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ABSTRACT

Nanoflare models of active region heating predict that coronal magnetic reconnection accelerates beams of nonthermal electrons, which deposit their energy in the transition region at the footpoints of hot loops, the moss. Beam heating brightens the moss for tens of seconds and blue-shifts its transition region lines, which heating by thermal conduction does not. Slit spectrographs have observed these signatures only when their slit happened to cross an event, since they cannot measure velocity across a wide field of view at a rapid cadence. A computed tomography imaging spectrograph (CTIS) can: it records several dispersed images of the same field at once, from which the intensity, Doppler shift, and width of an emission line can be recovered everywhere in a single exposure. We describe the second flight of the EUV Snapshot Imaging Spectrograph (ESIS-II), a sounding rocket CTIS which will observe an active region in Ne VII 465 Å and Si XII 499 Å, formed near 0.5 MK and 2 MK respectively, across a **XX** field of view every **XX**. Hydrodynamic simulations show that Ne VII discriminates between beam and conductive heating, while Si XII images extended coronal loops, the structures that are hardest for a CTIS to invert. ESIS-II reuses the cameras and avionics of the first flight, with new optics for the new passband: a primary mirror with a longer focal length, new gratings, and Si/Sc multilayer coatings, and two more channels for a total of six. We invert simulated ESIS-II observations with the multiplicative algebraic reconstruction technique using subsets of the six channels, and find that **XX** channels recover the Doppler shifts of extended structures to within **XX**. One sentence on what this means for the flight and for a future orbital CTIS.

Keywords: Solar coronal heating (1989), Solar magnetic reconnection (1504), Solar active regions (1974), Solar transition region (1532), Solar extreme ultraviolet emission (1493), Solar instruments (1499), Spectrometers (1554), Doppler shift (401)

1. INTRODUCTION

Outline.

- **Coronal heating by nanoflares.** Energy stored in the braided coronal field is released by many small reconnection events (E. N. Parker 1988). As in flares, much of it may go into nonthermal electrons which stream down the field and heat the transition region (TR) and chromosphere (A. G. Emslie 1978), and such beams arise in simulations of nanoflares (H. Bakke et al. 2018), so heating by electron beams is a signature of reconnection.

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- **The observational gap.** Beam heating shows up as brightenings and blue shifts in TR lines at the footpoints of hot loops, lasting tens of seconds (P. Testa et al. 2014). Velocities are measured by slit spectrographs such as the Interface Region Imaging Spectrograph (IRIS) (B. De Pontieu et al. 2014), whose field of view (FOV) and cadence catch only a few events, while imagers with a wide FOV measure no velocity at all.

- **Snapshot spectroscopy.** A computed tomography imaging spectrograph (CTIS) records dispersed images of a wide FOV at once (T. Okamoto & I. Yamaguchi 1991; M. Descour & E. Dereniak 1995). The Multi-Order Solar EUV Spectrograph (MOSES) (C. C. Kankelborg & R. J. Thomas 2001; J. L. Fox et al. 2010) and the EUV Snapshot Imaging Spectrograph (ESIS) (J. D. Parker et al. 2022) demonstrated the

technique in the extreme ultraviolet (EUV). ESIS-I flew on 2019 September 30 with four channels in O V 630 Å, and recovered the Doppler shifts of explosive events, compact sources which act as their own slit. Extended structures were harder to validate.

- **ESIS-II.** The same concept, with new optics for Ne VII and Si XII and six channels, to observe an active region (AR). Planned launch date and range.
- **Roadmap.** Section 2 motivates the passband, Section 3 describes the instrument and what has changed since ESIS-I, Section 4 inverts test data, and Section 5 concludes.

2. SCIENTIFIC MOTIVATION

2.1. *Electron Beam Heating in Active Region Moss*

Outline.

- Moss is the reticulated TR emission at the footpoints of the hot loops in AR cores (T. Berger et al. 1999). Its intensity is proportional to the pressure of the loop, so it is a sensitive gauge of coronal heating (P. C. H. Martens et al. 2000).
- Moss was long found to be steady (S. K. Antiochos et al. 2003; D. H. Brooks & H. P. Warren 2009): either the heating is quasi-steady, or the events were smaller or shorter than earlier instruments could resolve.
- The High-resolution Coronal Imager (Hi-C) (K. Kobayashi et al. 2014) found moss brightening on timescales of about 15 s, at the footpoints of loops which appeared shortly afterward in the hotter the Atmospheric Imaging Assembly (AIA) (J. R. Lemen et al. 2012) channels (P. Testa et al. 2013).
- P. Testa et al. (2014) combined IRIS, AIA, and models: TR brightenings lasted 20 s to 60 s, brightened by up to two orders of magnitude, and showed Si IV blue shifts on the few occasions the slit crossed an event. Beams deposit their energy below the height at which Si IV forms and blue-shift it, while conduction red-shifts it.
- The response depends on the beam parameters and on the initial state of the loop, and beams do not always blue-shift the line, but a blue shift is a clear sign of a beam (V. Polito et al. 2018). P. Testa et al. (2023) constrained the beam parameters by comparing observations with models.
- The best discriminator, TR velocity, is limited by the FOV and cadence of slit spectrographs.
- **Figure:** an AR in AIA, showing the moss and the hot core.

2.2. *Science and Technology Objectives*

Outline.

- **Science goal:** determine the role of magnetic reconnection in heating the corona.
- **Science Objective 1:** determine the frequency and spatial location of heating by electron beams, from intensity and velocity measured simultaneously over a wide FOV.
- **Science Objective 2:** determine the heating parameters (timescale, total energy, and low-energy cutoff of the beam) of every beam heating event, by comparison with hydrodynamic models (P. Testa et al. 2023).
- **Technology objective:** validate the inversion of two-dimensional extended structures observed with a CTIS. Compact events can be interpreted with as few as two channels (J. L. Fox 2011; J. D. Parker et al. 2022), but the inversion of extended structures has had no independent check. Recovering three moments of the line takes at least three channels, and each additional channel is a check on the inversion.

2.3. *Velocity over a Wide Field of View*

Outline.

- A slitless spectrograph elongates a Doppler-shifted source along its dispersion direction, which in a single channel cannot be told apart from a spatial structure lying along that direction.
- The elongation due to velocity rotates with the dispersion direction while spatial structure does not, so two channels dispersing in different directions separate them, and their difference shows Doppler shifts directly (J. D. Parker et al. 2022, Figures 9–11).
- Compact brightenings act as their own slit, so Science Objectives 1 and 2 can be closed without an inversion. Extended structures overlap in the dispersed images and must be inverted, which is what the technology objective tests.
- **Figure:** a cartoon of a compact, Doppler-shifted source and of an extended, stationary source, as seen by two channels, and their differences.

2.4. *Choice of Passband*

Outline.

- Ne VII 465.2 Å and Si XII 499.4 Å are an order of magnitude brighter than any other line between 440 Å to 520 Å, so their images do not overlap and suffer little contamination (Figure 1).
- Ne VII forms in the upper TR, near 0.5 MK, hotter than the Si IV line of P. Testa et al. (2014), near 0.08 MK (Figure 2). Si XII forms in the corona, near 2 MK, and images the extended loops needed for the technology objective.
- Does Ne VII respond to beams as Si IV does? HYDRAD (S. J. Bradshaw & P. J. Cargill 2013; J. W. Reep et al. 2013) simulations of a nanoflare whose energy is carried down by an electron beam, and of one whose energy is carried down by thermal conduction: Ne VII is blue-shifted early in the beam case and red-shifted in the conduction case, and its intensity decays more slowly in the beam case.

- Lines between 215 Å to 267 Å would fall on the passband in second order, so the coatings must suppress them (Section 3.3).
- **Figure:** the Ne VII intensity and line profile versus time for the beam and conduction simulations.

2.5. *Requirements*

Outline.

- **Table:** the science traceability matrix, from the objectives to the measurement and instrument requirements, and the performance expected from the ESIS-II model.
- **FOV and cadence:** at least the FOV of Hi-C (K. Kobayashi et al. 2014), which captured beam heating events, and a cadence at least as fast as its 2 s, since TR velocities can evolve faster than the 4 s Si IV cadence of P. Testa et al. (2014).
- **Spatial resolution:** a modest resolution suffices, since the brightenings grow by two orders of magnitude and the longer ones are visible at the resolution of AIA (P. Testa et al. 2014). Cadence and FOV, not resolution, have limited previous observations.
- **Dispersion:** enough to separate the FOV of Ne VII from that of Si XII on the detector.
- **Signal-to-noise ratio (SNR):** 100 photons to measure a velocity, spread over at least three channels, is 36 photons per channel per resolution element per exposure (an SNR of 6). With the radiances of J. E. Vernazza & E. M. Reeves (1978), this sets the required effective area.
- **Number of channels:** at least four, three for the moments and one to validate.
- **Mission:** an AR on the disk, at least 300 s of observing above 150 km, stable pointing, and recovery of the payload.

3. THE ESIS-II INSTRUMENT

Outline.

- A six-channel CTIS: a parabolic primary mirror images the Sun onto an octagonal field stop, which six gratings re-image onto six charge-coupled devices (CCDs), each channel dispersing in a different direction.
- Describe only what has changed. The heritage design is described in the ESIS-I instrument paper (Smart et al., in preparation). Cite once it is on arXiv.
- The design has changed considerably since the proposal. Every number comes from the current ESIS-II model, `esis.flights.f2`, in the `esis` package.

3.1. *Changes Since the First Flight*

Outline.

- **New:** the passband; a primary mirror with a longer focal length, which uses the full length of the optical table now that MOSES is gone from it; new gratings; Si/Sc multilayer coatings; two more channels; filters moved to suit the slower beam; the telemetry interface (Ethernet via telemetry (EVTM)).
- **Reused:** the cameras of ESIS-I, the data acquisition and control system (DACS), and the ground support equipment. What else carries over unchanged: the field stop mount, the grating mounts, the camera mount?
- **Table:** each subsystem of ESIS-I, and whether ESIS-II reuses, modifies, or replaces it.
- **Figure:** ESIS-I and ESIS-II on the optical table.

3.2. *Optical Design*

Outline.

- The primary mirror moves aft and the gratings forward on the optical table, lengthening the focal length of the primary for a finer plate scale. The field stop moves to the new focus.
- The radius, tilt, and varied line spacing (VLS) ruling of the gratings, first estimated from the grating equation and L. Poletto & R. J. Thomas (2004), then optimized for Ne VII and Si XII. The gratings are blazed.
- Ne VII and Si XII fall on opposite sides of the center of each detector without overlapping (Figure 3).
- The six channels take the azimuths of the six-channel layout for which ESIS-I was built.
- Plate scale, dispersion, and resolution from the ray-trace.
- **Table:** the optical prescription.
- **Figures:** the layout of the six channels and their dispersion directions; spot diagrams across the FOV at both wavelengths.

3.3. *Multilayer Coatings and Filters*

Outline.

- The primary mirror and the gratings both need multilayers which reflect Ne VII and Si XII well and suppress 215 Å to 267 Å, which would otherwise reach the detector in second order.
- Si/Sc multilayers on both, the coating the Solar Upper Transition Region Imager (SUTRI) flew for Ne VII (X. Bai et al. 2023; J. Wu et al. 2022), compared with the Al/Sc alternative (J. Rebellato et al. 2020, 2021), which reflects less with the same interfaces and number of periods, mostly at Si XII.
- Five periods, to keep the coating thin. More periods would reflect more, if a thicker coating is not more likely to fail.

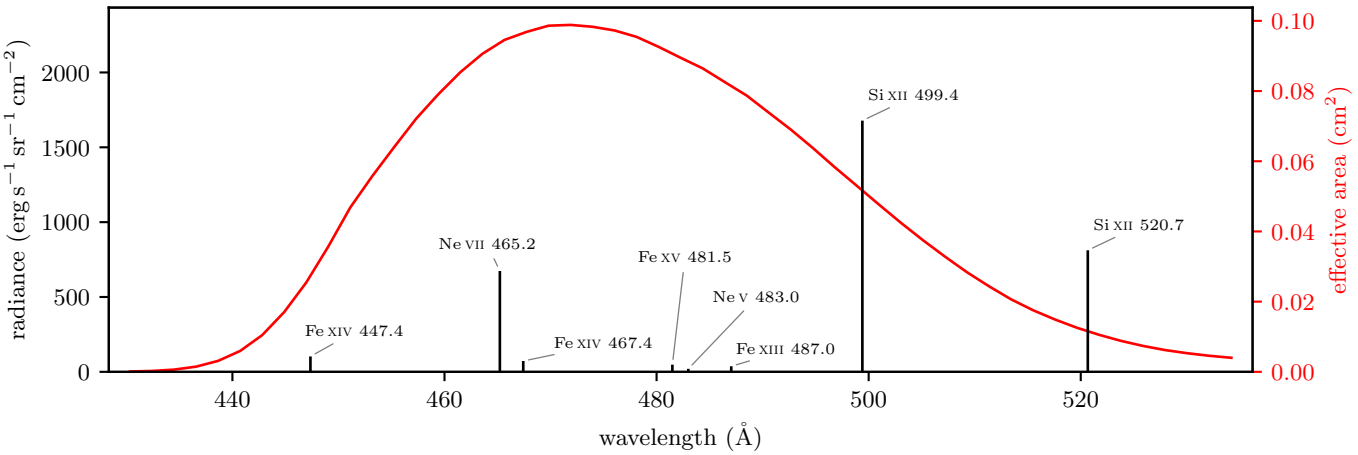


Figure 1. The eight emission lines which put the most light on the detector of ESIS-II, those whose radiance times the effective area is greatest, and the effective area of a single channel beneath them. The lines are computed from version 11.0.2 of the CHIANTI atomic database (K. P. Dere et al. 1997; G. Del Zanna et al. 2021), using the coronal abundances of `sun_coronal_2021_chianti` (K. P. Dere et al. 2023), which raise the photospheric abundances of M. Asplund et al. (2021) for the elements of low first ionization potential, the AR differential emission measure (DEM) file `active_region.dem`, and $n_eT = 3 \times 10^{15}$ K cm^{-3}.

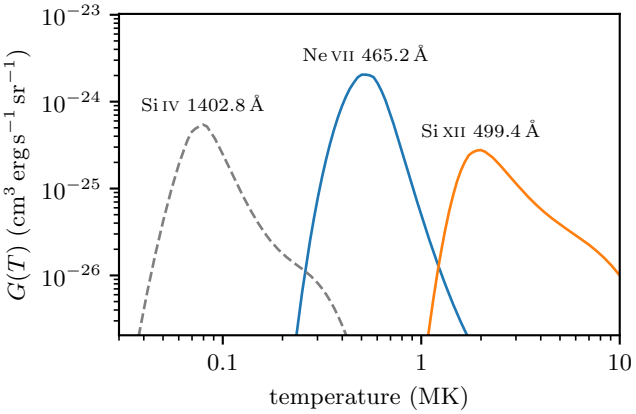


Figure 2. The contribution functions of the two lines ESIS-II observes (solid), and of the Si IV line in which P. Testa et al. (2014) saw the upflows of beam heating with IRIS (dashed). Each is the emissivity of the line per steradian divided by n_en_H, computed from version 11.0.2 of the CHIANTI atomic database (K. P. Dere et al. 1997; G. Del Zanna et al. 2021) with the abundances and the pressure of Figure 1. The peaks, at 0.08 MK, 0.5 MK, and 2 MK, are the temperatures at which the lines are formed.

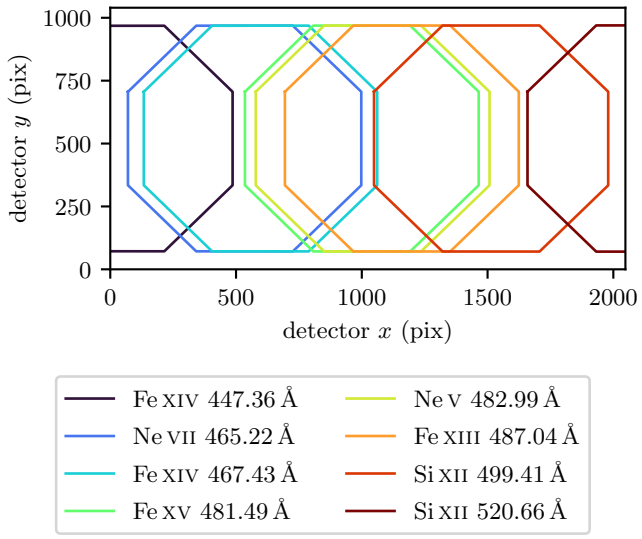


Figure 3. The image of the field stop on the detector of a single ESIS-II channel, at each of the eight lines of Figure 1. The axes span the active area of the CCD.

- The model of the coating follows a test sample of Si/Sc whose reflectance E. Gullikson of Lawrence Berkeley National Laboratory (LBNL) measured at beamline 6.3.2 of the Advanced Light Source: amorphous Si, a native oxide on the top layer, and the widths of the interfaces fitted to the measurement. The reflectance of witness samples of the flight coatings, once they are made.

- Filters: the ESIS-I design, with the transmission modeled including an oxide layer.

- **Figure:** the reflectance of the coatings and the transmission of the filters versus wavelength, and the measured reflectance of the test sample with the fit.

3.4. *Cameras and Avionics*

Outline.

- The cameras of ESIS-I and the new cameras which make up six. The readout sets the cadence, **XX**, with no dead time between exposures.

- Gain, read noise, and dark current of all six cameras. Measured with `msfc-ccd`.

- The DACS, with new cables for the new cameras and telemetry through EVTM.

3.5. *Mechanical Changes*

Outline.

- The new positions of the primary mirror, field stop, gratings, and filters on the optical table, and the effect on the sag of the table measured while aligning ESIS-I.
- The ESIS-I primary mirror shifted by about 10′ in vibration testing, and how the ESIS-II mount prevents that. Is the stiffer mount from the proposal still the plan?

3.6. *Alignment and Focus*

Outline.

- The ESIS-I procedure, with a visible-light alignment grating for each channel whose ruling is scaled by the ratio of the HeNe wavelength to the center of the passband, so that 632.8 nm light follows the path of the EUV.
- Establish the optic axis; align the primary mirror; place the field stop at its focus; align each grating to its CCD; focus each CCD; transfer the alignment from the visible gratings to the EUV gratings.
- As-measured alignment and focus, once the instrument is aligned.

3.7. *Effective Area and Expected Count Rates*

Outline.

- Effective area from the geometric area, the reflectance of the two coatings, the groove efficiency, the filter transmission, and the quantum efficiency (QE) of the CCD, compared with the requirement of Section 2.5.
- Expected count rates from an AIA DEM (J. Plowman & A. Caspi 2020) of an example AR (Figure 4), and the SNR of a single exposure across the FOV. The brightness of Ne VII depends on the DEM below $\log T = 5.8$, which AIA constrains poorly: this inversion gives Ne VII 2.4 times brighter than a demregpy DEM of the same images, while the Si XII lines agree to 7%. J. E. Vernazza & E. M. Reeves (1978) measured Ne VII/Si XII = 1.0 in an active region, against 0.8 from this inversion and 0.3 from demregpy, so this inversion is the closer of the two. Details in esis-mission/esis#72.
- **Figure:** the effective area versus wavelength.

3.8. *Concept of Operations*

Outline.

- Launch from White Sands Missile Range (WSMR), with the target AR chosen the day before.
- The cameras expose continuously every **XX**; frames taken before the door opens and after it closes are darks.
- Uplink commands can restart the cameras or repoint the payload. Only a subset of the frames is telemetered, so the payload must be recovered.

4. INVERSIONS OF TEST DATA

Outline.

- The test data are simulated ESIS-II observations of scenes whose line profiles are known, so the inversions can be compared with the truth.

4.1. *Test Data*

Outline.

- Scenes with known line profiles: IRIS Si IV rasters of ARs rescaled to Ne VII (B. De Pontieu et al. 2014); AIA images given the radiance and width of Si XII; and beam heating events from the HYDRAD simulations inserted into the moss. The first two are `esis.data.synth.scene_iris` and `scene_aia`.
- A radiative magnetohydrodynamic (MHD) simulation of an AR with MURaM (M. Rempel et al. 2023), whose velocity-resolved DEM gives the line profiles directly, with `esis.flights.f2.data.synth.scene_muram` (Figure 5).
- Forward modeled through `esis.flights.f2`, with photon and read noise for a single exposure.
- **Figure:** a simulated image from every channel.

4.2. *Inversion Method*

Outline.

- Multiplicative algebraic reconstruction technique (MART) (R. Gordon et al. 1970), as applied to ESIS-I (J. D. Parker et al. 2022, Appendix B): multiplicative updates keep the solution positive and assume no line shape, and regularization suppresses the "plaid" artifacts along the projection directions.
- Implemented in `ctis`.

4.3. *Validation Experiments*

Outline.

- Invert using every subset of three or more channels, and compare the recovered moments with each other and with the truth.
- Invert using a subset, forward model the withheld channels, and compare them with the simulated observations. Only this experiment can be repeated with the flight data, so the test data calibrate how well the withheld-channel residual predicts the true error.
- Error in intensity, Doppler shift, and width versus the number of channels, separately for compact brightenings in Ne VII and extended loops in Si XII.

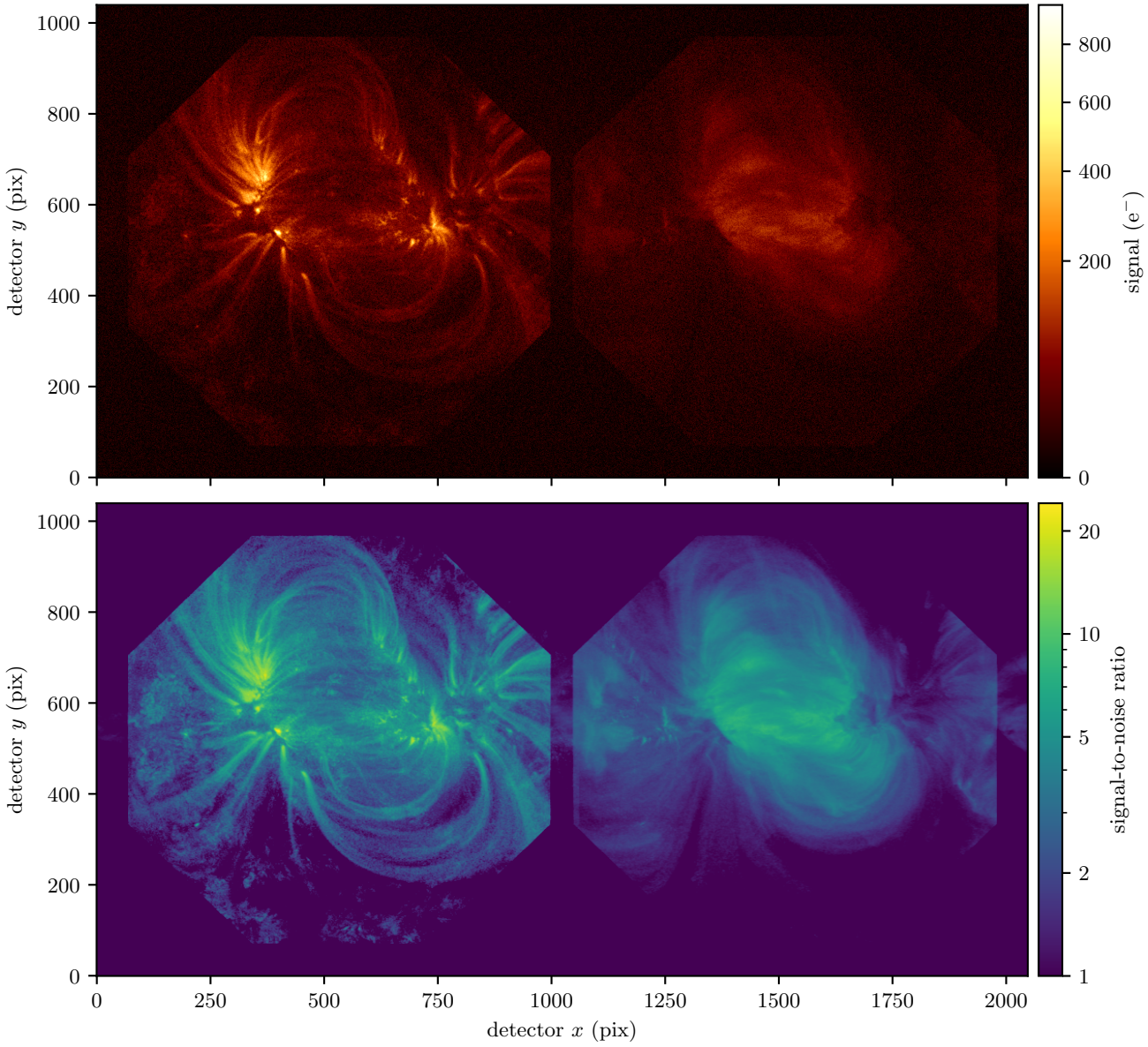


Figure 4. Above, a simulated 1.2 s exposure from one channel of ESIS-II, in the electrons collected by each pixel of the CCD, on a square-root scale, with the shot noise of the photons and the read noise of the camera. The scene is the active regions NOAA 12565 and 12567 at 00:00 UT on 2016 July 15: the DEM of each AIA pixel, inverted from the six optically thin EUV channels with the method of J. Plowman & A. Caspi (2020), and the twenty-two lines of the passband which carry at least 0.5% of its radiance, at rest. The images of Ne VII and Si XII fall on either side of the center. The fainter images around them are of Fe XIV, almost on top of Ne VII; of Fe XV, Ne V, and Fe XIII across the center; of a second line of Fe XIII and the weaker line of Si XII, which clip the right edge; and of shorter lines of Fe XIV, S XIV, and Mg IX, which clip the left edge. The lines of Mg VII and Mg VIII at the short end of the passband put almost no light on the detector, since the coatings reflect little there, and their images fall mostly off its left edge. Below, on a logarithmic scale, the SNR of each resolution element, a block of 2×2 pixels: the electrons it collects on average, over the standard deviation of the noise of their sum, from the shot noise of the photons it absorbs, the spread of the charge each frees, and the read noise of each of its pixels. It leaves out the charge which diffuses between pixels, which lowers the noise of each pixel only by correlating it with its neighbors.

4.4. *Results*

Outline.

- Compact Ne VII brightenings: whether the blue shifts of beam heating events are recovered with few channels, as for the explosive events of ESIS-I (J. D. Parker et al. 2022).
- Extended Si XII structures: **XX** channels recover the Doppler shift to within **XX**.

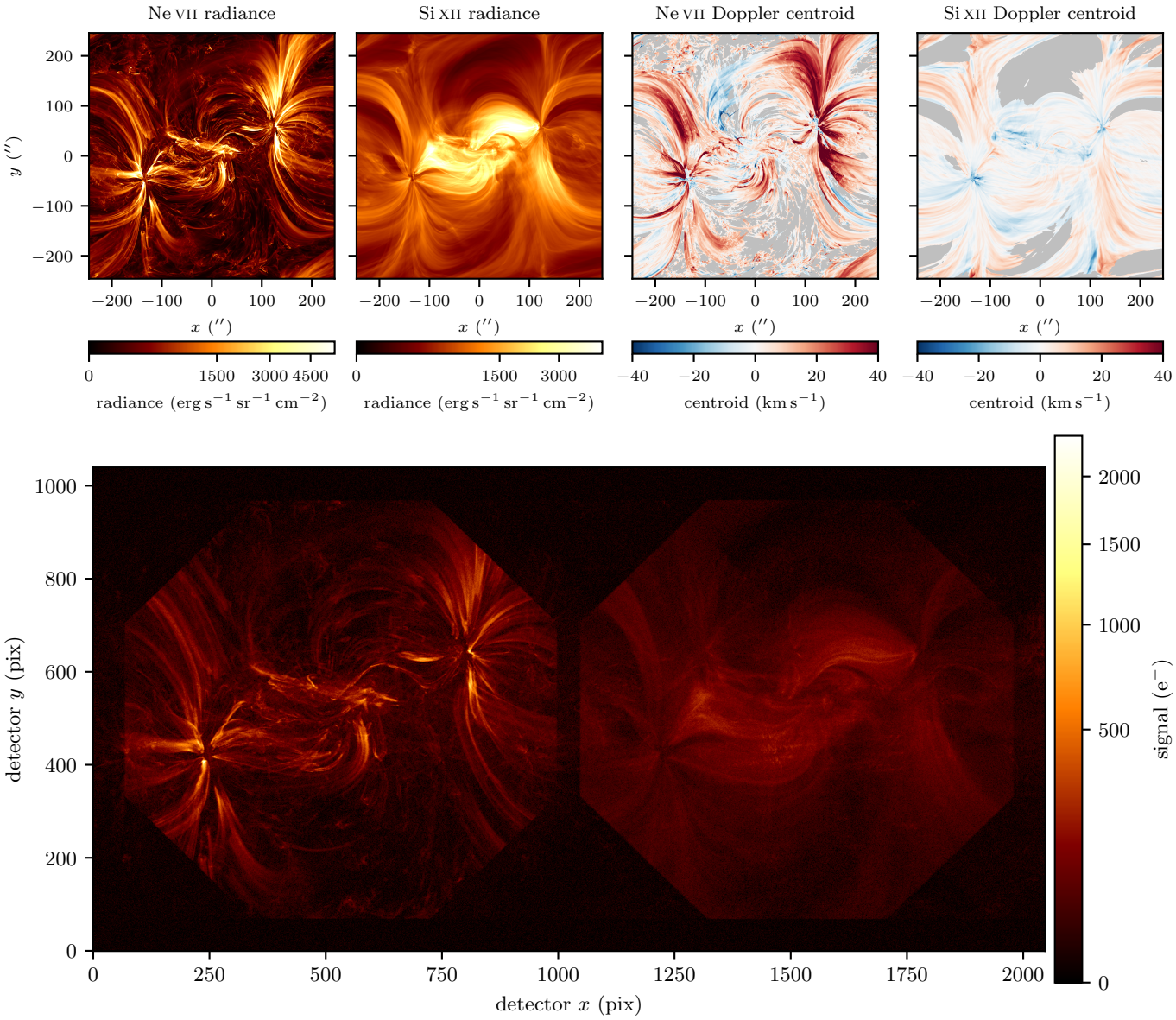


Figure 5. Above, a radiative MHD simulation of an active region with MURaM, before its M6.6 flare (M. Rempel et al. 2023): the radiance and the Doppler centroid of Ne VII and Si XII, positive away from the observer, synthesized from its velocity-resolved DEM. The plasma in each bin of temperature and of line-of-sight velocity emits a Gaussian at the center of the bin, with its thermal width and the width of plasma spread evenly across the bin. Each pixel of the simulation is stretched to span one pixel of the detector, so that the simulation fills the field of view at 1.8 times its true size. The Doppler centroid is not drawn, in gray, where the radiance is among the faintest 20% of its map. Below, a simulated 1.2 s exposure from one channel of ESIS-II of the twenty-five lines of the passband which carry at least 0.5% of its radiance, in the electrons collected by each pixel of the CCD, on a square-root scale, with the shot noise of the photons and the read noise of the camera.

- **Figures:** recovered versus true moments; error versus the number of channels.

5. CONCLUSIONS

Outline.

- How the Ne VII and Si XII passband serves the two science objectives and the technology objective.
- What changed since ESIS-I, and the readiness of the instrument for flight.
- The minimum number of channels found by the inversions, and what it implies for a future orbital CTIS.
- The data products, Levels 0 through 3, and the software which makes them, released through the `esis` package and the Solar Data Analysis Center.

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